Imperial College London

Department of Electronic and Electrical Engineering

Examinations 2025

Digital Signal Processing and Digital Filters

MSc and EEE/EIE PART IV

April, 2025 • Exam Duration: Three Hours

There are FOUR questions on this paper.

Answer Question 1 and any TWO other questions.

Question 1 is worth 40% of the total marks
while each of the other questions is worth 30%.

If necessary, make reasonable assumptions for your solutions and clearly explain them in your answers.

A formula sheet is attached in the next two pages.

Markers: Ayush Bhandari and Tania Stathaki.

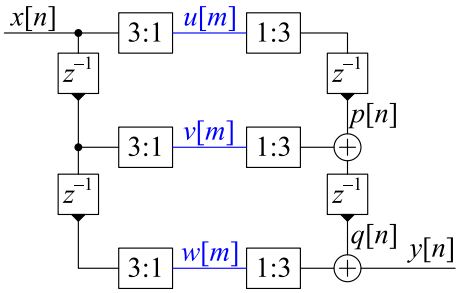
Question 1 [40 Marks]

Q1-1 **Multirate Systems.**

A certain multirate DSP system is described in the block diagram. Given that the output of this system is:

$$y[n] = [\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}, \mathtt{e}, \mathtt{f}, \mathtt{g}, \mathtt{h}, \mathtt{i}].$$

Identify each of the signals:
$x[n], u[m], p[n], v[m], q[n], w[m]$.

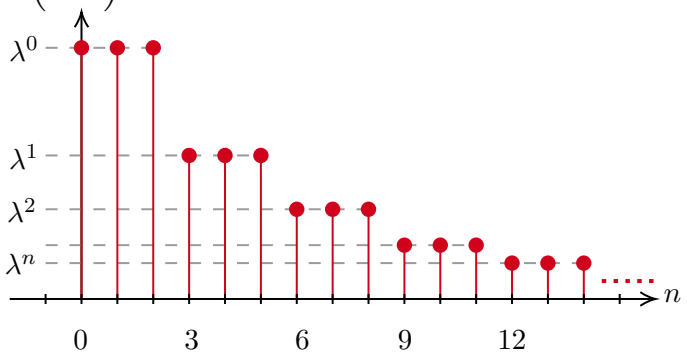


Notation: $y[n] = [\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}, \mathtt{e}, \mathtt{f}]$ means that $y[0] = \mathtt{a}$, $y[1] = \mathtt{b}$ and so on, and that $y[n] = 0$ outside this range. **[6 Marks]**

$x[n]$	cdefghijklmn
$u[m]$	c f i l
$p[n]$	-c--f--i--l
$v[m]$	b e h k
$q[n]$	-bc-ef-hi-kl
$w[m]$	a d g j

Q1-2 The impulse response of a certain filter is shown below.

$$h_\lambda[n] = \left(\lambda^{\lfloor \frac{n}{3} \rfloor}\right) u[n],\ \lambda < 1$$



(a) Compute the z-transform, $H_\lambda(z)$ of $h_\lambda[n] = \left(\lambda^{\lfloor n/3 \rfloor}\right) u[n]$, $\lambda < 1$. **[4 Marks]**

The z-transform of the sequence $h_\lambda[n]$ is defined as

$$\begin{aligned} H_\lambda(z) &= \sum_{n=-\infty}^{\infty} h_\lambda[n] z^{-n} \\ &= \sum_{n=0}^{\infty} \lambda^{\lfloor n/3 \rfloor} z^{-n} = \sum_{k=0}^{\infty} \lambda^k z^{-3k} + \sum_{k=0}^{\infty} \lambda^k z^{-3k-1} + \sum_{k=0}^{\infty} \lambda^k z^{-3k-2} \\ &= \frac{1 + z^{-1} + z^{-2}}{1 - \lambda z^{-3}}. \end{aligned}$$

(b) Show that when $\lambda = 0$, the z-transform, $H_0(z)$ reduces to, $\left(1 + z + z^2\right)/z^2$. **[2 Marks]**

From the expression, we know that

$$H_0(z) = 1 + z^{-1} + z^{-2}.$$

(c) Compute the impulse response of $H_0(z)$. **[1 Mark]**

$h_0[n] = \delta[n] + \delta[n-1] + \delta[n-2]$.

Q1-3 **Continuous to Digital Filter.** Let $H(s)$ denote a second order continuous-time filter:

$$H\left(s\right) = \frac{1}{s^2 + \frac{s}{2} + 1}.$$

Obtain the corresponding digital filter $H_d(\omega)$.

Use the bilinear mapping $s = \frac{(z+1)}{(z-1)}$. You can assume the gain to be unity. **[3 Marks]**

Straight-forward application of the bilinear mapping implies that we have,

$$H\left(\frac{z+1}{z-1}\right) = \frac{2(z-1)^2}{5z^2 + 3}.$$

Q1-4 Let $\{x\}$ denote the fractional part of x, that is, $\{2.01\} = 0.1$ and so on.

(a) Plot $f(x)$ for the finite interval $0 \leqslant x \leqslant 3$. **[1 Marks]**

(b) Compute the Fourier transform of the function $f(x) = \{x\}$. **[4 Marks]**

Note that, f is a periodic function with periodicity of 1, hence

$$f(x) = \sum_{k=-\infty}^{\infty} a_k e^{\jmath 2\pi k x}$$

where the coefficients can be found via

$$a_k = \int_0^1 f(x) e^{-\jmath 2\pi k x} dx = \frac{1}{-\jmath 2\pi k} \int_0^1 x d(e^{-\jmath 2\pi k x}) = \frac{1}{-\jmath 2\pi k}(x e^{-\jmath 2\pi k x}\Big|_0^1 - \int_0^1 e^{-\jmath 2\pi k x} dx)$$
$$= \frac{1}{-\jmath 2\pi k}(e^{-\jmath 2\pi k} - \frac{1}{-\jmath 2\pi k}(e^{-\jmath 2\pi k} - 1)).$$

This gives the Fourier series expression.

Q1-5 Given z-transform

$$H\left(z\right) = \frac{2 + 2z^{-1}}{-5 + 0.5z^{-1} + z^{-2}}$$

(a) Draw the zero-pole plot. **[2 Marks]**.

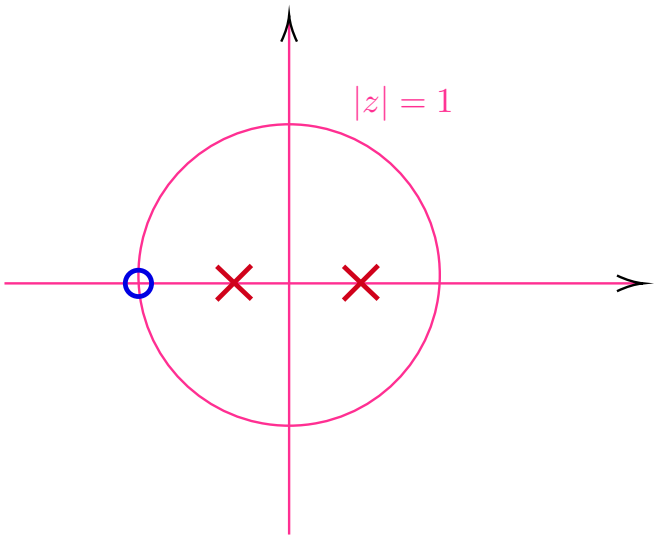
(b) Compute $|H\left(e^{\jmath\omega}\right)|$ at $\omega = 1.2$. **[2 Marks]**.

(c) Assuming that the system is causal, compute its impulse response. **[3 Marks]**.

(a) Notice that, by factor decomposition, we have

$$H(z) = \frac{2(1+z^{-1})}{(z^{-1}+2.5)(z^{-1}-2)}.$$

Hence, the zeros is -1 (marked in blue circle) and the poles are -0.4 and 0.5 (marked in red cross), which is plotted below.



(b) $|H(e^{j1.2})| = \frac{2|1+e^{-j1.2}|}{|2.5+e^{-j1.2}||e^{-j1.2}-2|} = 0.58.$

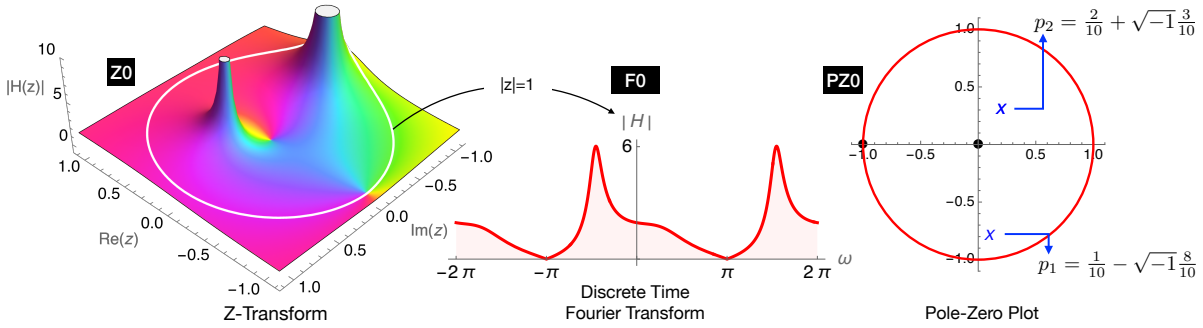
(c) From the fractional expansion, we have that

$$H(z) = \frac{2(1+z^{-1})}{(z^{-1}+2.5)(z^{-1}-2)} = \frac{-2}{7.5} \times \frac{1}{1-(-0.4)z^{-1}} + \frac{-2}{3} \times \frac{1}{1-0.5z^{-1}}.$$

Given that the system is causal, then the time-domain impulse response is given by

$$h[n] = \frac{-2}{7.5}(-0.4)^n u[n] + \frac{-2}{3}0.5^n u[n].$$

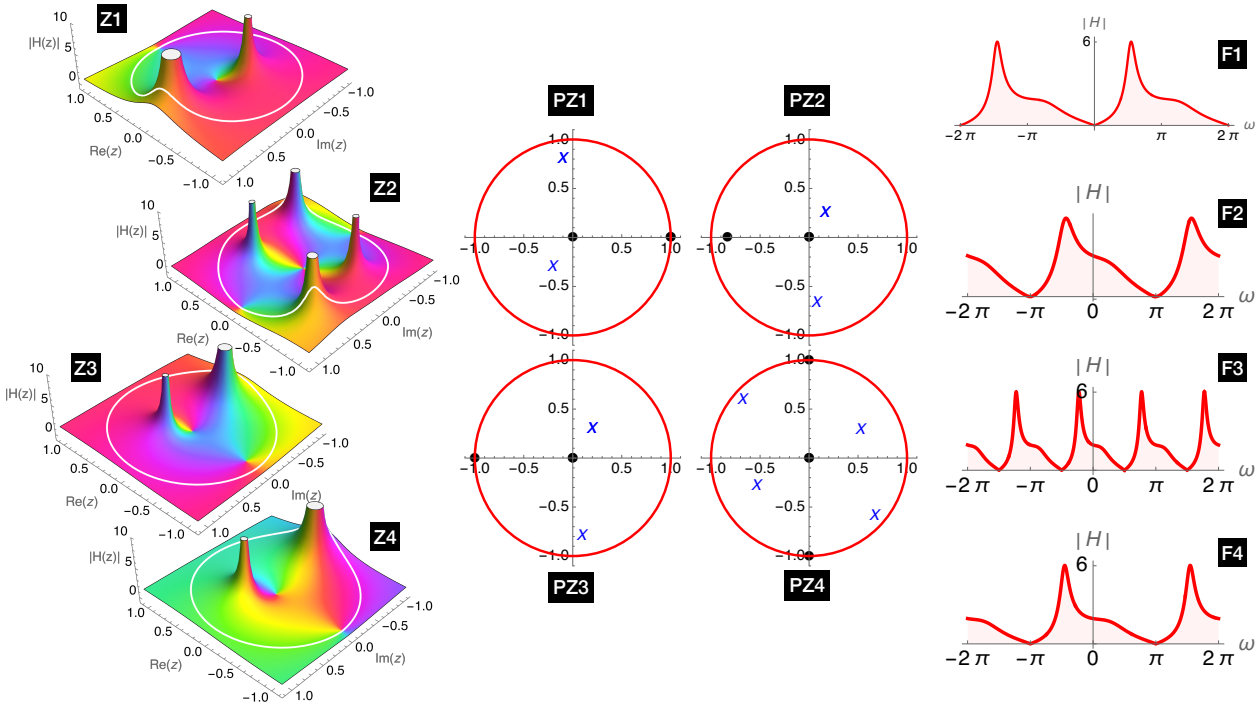
Q1-6 For a certain transfer function, its z-transform, discrete time Fourier transform (DTFT) and pole-zero plot are given by:



In the above, Z0-PZ0-F0 form a matching triplet.

Following this, from the figure below, match the z-transforms (Z1-Z4) and DTFT (F1-F4) with their respective pole-zero plots (PZ1-PZ4). Provide a brief reasoning for your answer.

[8 Marks]



Z1-PZ1-F1 Z2-PZ4-F3 Z3-PZ2-F2 Z4-PZ3-F4.

This matching can be found by checking: (1) the number of roots; (2) the number of copies in spectrum; (3) distribution of the poles and (4) the distance between the unit length contour and poles.

Q1-7 A causal sequence $f[n]$ with *non-minimum phase* property has a z-transform:

$$F(z) = \frac{\prod_{m=1}^{m=4}\left(1 - f_m z^{-1}\right)}{\left(1 - z^{-1}\right)^2\left(1 - \frac{1}{8}z^{-1}\right)} \quad \text{where} \quad \begin{bmatrix} f_1 & f_2 \\ f_3 & f_4 \end{bmatrix} = \begin{bmatrix} -3/2 & 1/4 \\ 1/3 & -5/2 \end{bmatrix}.$$

Determine the conditions on μ so that the sequence $g[n] = \mu^n f[n]$ is a *minimum phase* sequence.

For $g[n]$ to satisfy minimum phase property, all of its poles/zeros should be inside the unit circle.

[4 Marks]

Clearly, $G(z) = F(z/\mu)$. Note that all of the poles of $F(z)$ are inside the unit circle. The only zeros that live in the exterior of $|z| > 1$ are f_1 and f_4. Hence the largest potential zero of $Y(z)$ has the value $5\mu/2$. Consequently, $|\mu| < 2/5$ guarantees minimum phase property.

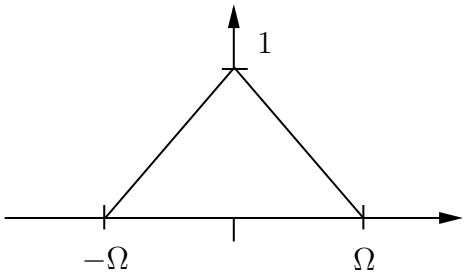


Fig. 1: $X(e^{\jmath\omega}) : \Omega < \pi$.

Question 2 [30 Marks]

Q2-1 The sequence $y[n]$ is the output of the system,

$$x[n] \rightarrow \boxed{1:K} \rightarrow v[n] \rightarrow \boxed{Q:1} \rightarrow y[n]$$

to the input sequence $x[n]$.

(a) Write down the expression for sequences $v[n]$ and $y[n]$ in terms of $x[n]$. **[2 Marks]**

$$v[n] = \begin{cases} x[\frac{n}{K}], & \text{if } n = \mathbb{Z}K \\ 0, & \text{otherwise} \end{cases}, \quad y[n] = v[Qn] = \begin{cases} x[\frac{nQ}{K}], & \text{if } n = \mathbb{Z}K \\ 0, & \text{otherwise} \end{cases}.$$

(b) Compute the Discrete-time Fourier transform (DTFT) of $v[n]$ and $y[n]$. **[2 Marks]**

$$V(e^{\jmath\omega}) = \sum_{n=-\infty}^{\infty} v[n]e^{-\jmath\omega n} = \sum_{m=-\infty}^{\infty} x[m]e^{-\jmath\omega mK} = X(e^{\jmath\omega K}),$$

$$Y(e^{\jmath\omega}) = \frac{1}{Q}\sum_{k=0}^{Q-1} V(e^{\frac{-\jmath 2\pi k}{Q}} e^{\frac{\jmath\omega}{Q}}) = \frac{1}{Q}\sum_{k=0}^{Q-1} X(e^{\frac{-\jmath 2\pi k}{Q}} e^{\frac{\jmath\omega K}{Q}}).$$

(c) Given that the input $x[n]$ is bandlimited with its DTFT shown in Fig. 1.

i. Let $K = 3$ and $Q = 2$. In order to reconstruct the waveform of the input $x[n]$, we need to low-pass filter the output $y[n]$. Compute the cut-off frequency of this low-pass filter. **[2 Marks]**

Notice that, the maximum frequency is $\frac{\Omega Q}{K} = \frac{2\Omega}{3}$. Hence, for any cut-off frequency lying in the range $[\frac{2\Omega}{3}, \frac{2\pi}{3}]$ can achieve the goal of signal reconstruction.

ii. Given $K = 3$ and the DTFT of $x[n]$ characterized in Fig. 1, compute the maximum downsampling factor such that there is no spectrum aliasing in y. **[2 Marks]**

The resulting maximum frequency $\frac{\Omega Q}{3} \leqslant \pi$ will avoid aliasing, leading to $Q \leqslant \frac{3\pi}{\Omega}$. Then the maximum downsampling factor is $\lfloor \frac{3\pi}{\Omega} \rfloor$.

Q2-2 A DSP system is depicted in Fig. 2.

(a) Write down the time-domain relation between input and output. **[3 Marks]**

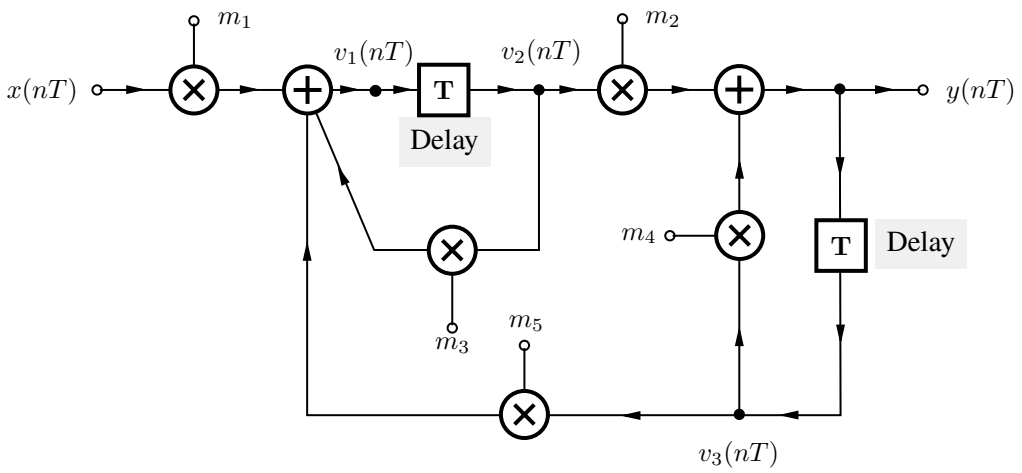


Fig. 2: System Model for Q2-2. T denotes delay by sample step T, $\otimes$ denotes *gain* or *multiplication* and $\oplus$ denotes *addition*.

From the figure, we know

$$v_1(nT) = m_1 x(nT) + m_3 v_2(nT) + m_5 v_3(nT),$$
$$v_2(nT) = \varepsilon^{-1} v_1(nT), \quad v_3(nT) = \varepsilon^{-1} y(nT)$$
$$y(nT) = m_2 v_2(nT) + m_4 v_3(nT)$$

and on eliminating $v_2(nT)$ and $v_3(nT)$ in $v_1(nT)$ and $y(nT)$, we have

$$(1 - m_3\varepsilon^{-1}) v_1(nT) = m_1 x(nT) + m_5 \varepsilon^{-1} y(nT)$$

and $(1 - m_4\varepsilon^{-1}) y(nT) = m_2 \varepsilon^{-1} v_1(nT)$. Therefore, we have

$$y(nT) = a_1 x(nT - T) + b_1 y(nT - T) + b_2 y(nT - 2T),$$

where $a_1 = m_1 m_2$, $b_1 = m_3 + m_4$ and $b_2 = m_2 m_5 - m_3 m_4$.

(b) Write down the frequency response of this system. **[3 Marks]**

Let $x[n] = x(nT)$ and $y[n] = y(nT)$. Then, the time-domain difference equation can be written as

$$y[n] - b_1 y[n-1] - b_2 y[n-2] = a_1 x[n-1].$$

Performing z-transform leads to

$$Y(z)(1 - b_1 z^{-1} - b_2 z^{-2}) = a_1 z^{-1} X(z) \Longrightarrow \frac{Y(z)}{X(z)} = \frac{a_1 z^{-1}}{1 - b_1 z^{-1} - b_2 z^{-2}}.$$

(c) Compute the zeros and poles of the filter system. **[2 Marks]**

The zero lies in $z = 0$ and poles are the roots of the quadratic equation $z^2 - b_1 z - b_2 = 0$. Namely, the poles are $\frac{b_1 \pm \sqrt{b_1^2 + 4b_2}}{2}$.

(d) Identify the conditions for the system to be stable. **[2 Marks]**

The poles are within the unit circle, $|\frac{b_1 \pm \sqrt{b_1^2 + 4b_2}}{2}| < 1$.

Q2-3 The input-output relation of a system is specified by,

$$\sum_{r=0}^{M} a[r]\, y[n-r] = \sum_{r=0}^{M} b[r]\, x[n-r],$$

where $b[n] = a^*[M-n]$.

(a) Show that the magnitude response of the transfer function is unity. **[3 Marks]**

The transfer function is given by

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{r=0}^{M} a^*[M-r]\, z^{-r}}{\sum_{r=0}^{M} a[r]\, z^{-r}} = z^{-M} \frac{\sum_{r=0}^{M} a^*[r]\, z^{r}}{\sum_{r=0}^{M} a[r]\, z^{-r}}$$

. Hence, we know that $|H(e^{\jmath\omega})| = |e^{-\jmath\omega M}||\frac{\sum_{r=0}^{M} a^*[r]z^r}{\sum_{r=0}^{M} a[r]z^{-r}}| = 1$.

(b) Write down z-transform $H(z)$ for 1st-order filter ($M = 1$) with $a[n] \in \mathbb{R}$. **[2 Marks]**

$H(z) = \frac{a^*[1]+a^*[0]z^{-1}}{a[0]+a[1]z^{-1}} = \frac{z^{-1}+p}{pz^{-1}+1}$ where $p = a[1]/a[0]$.

(c) Show that the group delay for the filter,

$$H(z) = \frac{z^{-1} - \alpha^*}{1 - \alpha z^{-1}}$$

is always non-negative. **[7 Marks]**

Show that the group delay of an allpass filter is nonnegative for all ω.

If α is real and $|\alpha| < 1$, the group delay of a filter that has a system function

$$H_1(z) = \frac{1}{1 - \alpha z^{-1}}$$

which has a single pole at $z = \alpha$, is

$$\tau_1(\omega) = -\frac{\alpha^2 - \alpha\cos\omega}{1 + \alpha^2 - 2\alpha\cos\omega}.$$

Similarly, the group delay for a filter with the system function $H_2(z) = 1 - \alpha z^{-1}$ which has a single zero at $z = \alpha$, is $\tau_2(\omega) = -\tau_1(\omega)$. Furthermore, if

$$H_3(z) = z^{-1} - \alpha = -\alpha\left[1 - \frac{1}{\alpha}z^{-1}\right],$$

the group delay is

$$\tau_3(\omega) = \frac{\alpha^{-2} - \alpha^{-1}\cos\omega}{1 + \alpha^{-2} - 2\alpha^{-1}\cos\omega} = \frac{1 - \alpha\cos\omega}{1 + \alpha^2 - 2\alpha\cos\omega}.$$

Therefore, the group delay of a single all-pass factor of the form $H_0(z) = \frac{z^{-1}-\alpha}{1-\alpha z^{-1}}$ is

$$\begin{aligned}\tau_0(\omega) = \tau_1(\omega) + \tau_3(\omega) &= -\frac{\alpha^2 - \alpha\cos\omega}{1 + \alpha^2 - 2\alpha\cos\omega} + \frac{1 - \alpha\cos\omega}{1 + \alpha^2 - 2\alpha\cos\omega} \\ &= \frac{1 - \alpha^2}{1 + \alpha^2 - 2\alpha\cos\omega} = \frac{1 - \alpha^2}{|1 - \alpha e^{-j\omega}|^2}\end{aligned}$$

which, because $|\alpha| < 1$, is positive for all ω. For complex roots, the allpass factors have the form $H(z) = \frac{z^{-1}-\alpha^*}{1-\alpha z^{-1}}$. Therefore, with $\alpha = |\alpha|e^{j\theta}$, the frequency response is

$$H(e^{j\omega}) = \frac{e^{-j\omega} - |\alpha|e^{-j\theta}}{1 - |\alpha|e^{j\theta}e^{-j\omega}} = e^{-j\theta}\frac{e^{-j(\omega-\theta)} - |\alpha|}{1 - |\alpha|e^{-j(\omega-\theta)}}$$

and the group delay is

$$\tau(\omega) = \tau_1(\omega - \theta) = \frac{1 - \alpha^2}{|1 - \alpha e^{-j(\omega-\theta)}|^2}$$

which is nonnegative for all ω.

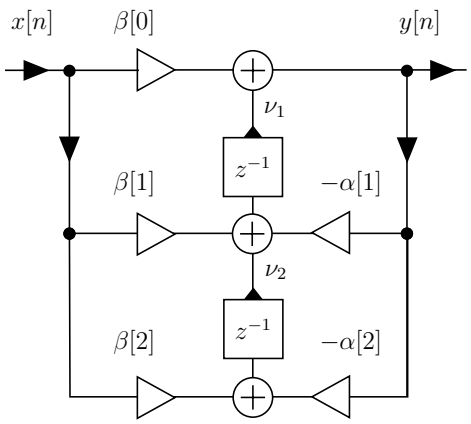


Fig. 3: Filter structure.

Question 3 [30 Marks]

Q3-1 A filtering system is characterized by the block diagram shown in Fig. 3.

(a) Write down the difference equation describing the relationship between $x[n]$ and $y[n]$. **[3 Marks]**

$\sum_{m=0}^{2} \alpha[m]y[n-m] = \sum_{m=0}^{2} \beta[m]x[n-m]$ where $\alpha[0] = 1$.

(b) Compute the transfer function $H(z)$ of this filter. **[3 Marks]**

The transfer function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{m=0}^{2} \beta[m]z^{-m}}{\sum_{m=0}^{2} \alpha[m]z^{-m}}.$$

(c) Find the conditions that make the system stable. **[2 Marks]**

The poles of the system are given by $\frac{-\alpha[1]\pm\sqrt{a^2[1]-4a[2]}}{2}$.

Then, if $|\frac{-\alpha[1]\pm\sqrt{a^2[1]-4a[2]}}{2}| < 1$, the system is stable.

Q3-2 Consider the system with the following impulse response

$$h\left[n\right] = \left(\mu\right)^{n+\pi} u\left[n-K\right],$$

where $u\left[n\right]$ denotes the step function and $\mu < 1$.

(a) Derive the phase of the system. **[3 Marks]**

(b) Derive the expression of the group delay $\tau_H\left(e^{\jmath\omega}\right)$. **[4 Marks]**

(c) Find μ, K such that $\tau_H\left(e^{\jmath\pi}\right) = 3$ and $\tau_H\left(e^{\jmath 0}\right) = 1$. **[5 Marks]**

i) The group delay satisfies $\tau_H\left(e^{\jmath\omega}\right) = -\frac{d}{d\omega}\angle H\left(e^{\jmath\omega}\right)$. We proceed by computing the frequency response function $H\left(e^{\jmath\omega}\right)$ as

$$H\left(e^{\jmath\omega}\right) = e^{-K\jmath\omega}\mu^{\pi}\frac{1}{1-\mu e^{-\jmath\omega}}.$$

Then

$$\angle H\left(e^{\jmath\omega}\right) = -K\omega - \angle\left(1-\mu e^{-\jmath\omega}\right) = -K\omega - \tan^{-1}\left(\frac{\mu\sin\omega}{1-\mu\cos\omega}\right)$$

ii) The group delay thus satisfies

$$\begin{aligned}\tau_H\left(e^{\jmath\omega}\right) &= -K - \frac{1}{1+\frac{\mu^\pi \sin^2\omega}{(1-\mu\cos\omega)^2}} \cdot \frac{\mu\cos\omega\left(1-\mu\cos\omega\right) - \mu\sin\omega\cdot\mu\sin\omega}{\left(1-\mu\cos\omega\right)^2} \\ &= -K - \frac{\mu\cos\omega - \mu^\pi\cos^2\omega - \mu^\pi\sin^2\omega}{1+\mu^\pi\cos^2\omega - 2\mu\cos\omega + \mu^\pi\sin^2\omega} \\ &= -K - \frac{\mu\cos\omega - \mu^\pi}{1-2\mu\cos\omega+\mu^\pi}\end{aligned}$$

iii) $\tau_H\left(e^{\jmath\pi}\right) = -K - \frac{-\frac{1}{\mu}-1}{\frac{1}{\mu^\pi}+2\frac{1}{\mu}+1} = -K + \frac{1}{\frac{1}{\mu}+1} = 3.$

$\tau_H\left(e^{\jmath 0}\right) = -K - \frac{\frac{1}{\mu}-1}{\frac{1}{\mu^\pi}-2\frac{1}{\mu}+1} = -K - \frac{1}{\frac{1}{\mu}-1} = 1.$

Using the two equations above we get two solutions $\mu = -1.61, K = -0.38$ and $\mu = 0.62, K = -2.62$.

Q3-3 **Multirate System**

Consider the following system with input $x\left[n\right]$ and output $y\left[n\right]$.

$$x\left[n\right] \rightarrow \boxed{1:3} \rightarrow \boxed{2:1} \rightarrow \boxed{\frac{z^{-12}}{\alpha - z^{-12} + \beta z^{-24}}} \rightarrow \boxed{2:1} \rightarrow \boxed{1:3} \rightarrow y\left[n\right]$$

Suppose that an equivalent representation of the above system is given by,

$$x\left[n\right] \rightarrow \boxed{4:1} \rightarrow \boxed{\frac{B\left(z\right)}{A\left(z\right)}} \rightarrow \boxed{1:9} \rightarrow y\left[n\right].$$

Identify α and β if $A\left(z\right)$ has roots at $1/2$ and $1/3$. **[10 Marks]**

The following sequence leads to the solution,

$$x\left[n\right] \rightarrow \underbrace{\boxed{1:3} \rightarrow \boxed{2:1}}_{\text{Exchange}} \rightarrow \underbrace{\boxed{H\left(z\right)} \rightarrow \boxed{2:1}}_{\text{Use Noble Identity}} \rightarrow \boxed{1:3} \rightarrow y\left[n\right]$$

$$x\left[n\right] \rightarrow \boxed{2:1} \rightarrow \underbrace{\boxed{1:3} \rightarrow \boxed{2:1}}_{\text{Exchange}} \rightarrow \boxed{H_1\left(z\right)} \rightarrow \boxed{1:3} \rightarrow y\left[n\right]$$

$$x\left[n\right] \rightarrow \underbrace{\boxed{2:1} \rightarrow \boxed{2:1}}_{\text{Merge}} \rightarrow \underbrace{\boxed{1:3} \rightarrow \boxed{H_1\left(z\right)}}_{\text{Exchange using Noble Identity}} \rightarrow \boxed{1:3} \rightarrow y\left[n\right]$$

$$x\left[n\right] \rightarrow \boxed{4:1} \rightarrow \boxed{H_2\left(z\right)} \rightarrow \underbrace{\boxed{1:3} \rightarrow \boxed{1:3}}_{\text{Merge}} \rightarrow y\left[n\right]$$

$$x\left[n\right] \rightarrow \boxed{4:1} \rightarrow \boxed{H_2\left(z\right)} \rightarrow \boxed{1:9} \rightarrow y\left[n\right]$$

where,

$$H_1\left(z\right) = \frac{z^{-6}}{\alpha - z^{-6} + \beta z^{-12}} \quad \text{and} \quad H_2\left(z\right) = \frac{z^{-2}}{\alpha - z^{-2} + \beta z^{-4}}$$

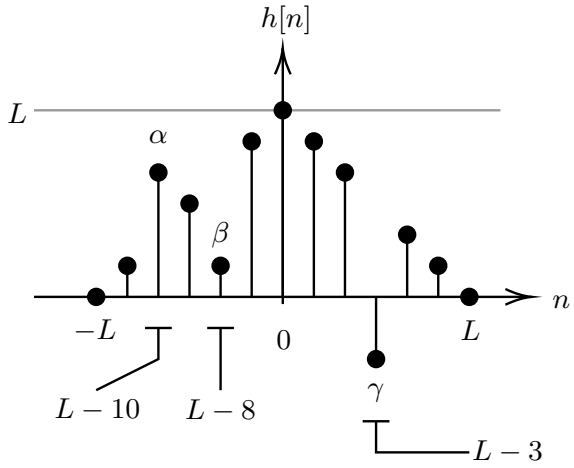
yielding, $B\left(z\right) = z^{-2}$ and $A\left(z\right) = \alpha - z^{-2} + \beta z^{-4}$. We are given that,

$$A\left(\frac{1}{3}\right) = 0 \Rightarrow \alpha - 9 + 81\beta = 0 \text{ and } A\left(\frac{1}{2}\right) = 0 \Rightarrow \alpha - 4 + 16\beta = 0$$

and hence, $\alpha = 36/13$ and $\beta = 1/13$.

Question 4 [30 Marks]

Q4-1 Let $h[n]$ be the filter given by,



while it is given that $L = 6$.

(a) Let $g[n] = d_2 * h[n]$, where $d_2 = d_1 * d_1$, and $d_1[n] = \delta[n] - \delta[n-1]$. Find α, β, γ so that,

$$g[n] = \delta[n].$$

[3 Marks]

(b) Derive the expression of $G(j\omega)$, the DTFT of $g[n]$. **[3 Marks]**

(c) Find the analytical relationship between $G(j\omega)$ and $H(j\omega)$, the DTFT of $h[n]$. **[3 Marks]**

(d) Using the convolution theorem for the DTFT evaluate $H(j\omega)$. **[3 Marks]**

i) We have that

$$d_2[n] = d_1[n] * d_1[n] = \delta[n] - 2\delta[n-1] + \delta[n-2]. \quad (1)$$

Then,

$$g[n] = \sum_{k=-\infty}^{\infty} d_2[k]\, h[n-k] = d_2[0]\, h[n-0] - 2d_2[1]\, h[n-1] + d_2[2]\, h[n-2] \quad (2)$$

$$= h[n] - 2h[n-1] + h[n-2] = h[n] - h[n-1] - h[n-1] + h[n-2]. \quad (3)$$

The constraint $g[n] = \delta[n]$ implies that $g[L-10] = 0$. Using $h[n] = L - |n|$ we get:

$$g[L-10] = h[L-10] - 2h[L-11] + h[L-12] = \alpha - 2 \cdot 1 + 0. \quad (4)$$

It can be seen that (4) will be true only for $\alpha = 2$. Similarly, we can write for β and γ:

$$g[L-8] = h[L-8] - 2h[L-9] + h[L-10] = \beta - 2 \cdot 3 + \alpha \quad (5)$$

$$g[L-3] = h[L-3] - 2h[L-4] + h[L-5] = \gamma - 2 \cdot 4 + 5 = \gamma - 3. \quad (6)$$

Given that $\alpha = 2$ it follows that $\beta = 4$ and $\gamma = 3$. In conclusion, the parameters $\alpha = 2, \beta = 4, \gamma = 3$ satisfies $g[n] = \delta[n]$.

ii) $G(j\omega)$ satisfies

$$G(j\omega) = \sum_{n=-\infty}^{\infty} \delta[n]\, e^{-j\omega n} = 1. \tag{7}$$

iii) Let $D_2(j\omega)$ denote the DTFT of $d_2[n]$. Then, according the the convolution theorem,

$$G(j\omega) = H(j\omega)\cdot D_2(j\omega) = H(j\omega)\cdot\left(e^{-j\omega\cdot 0} - 2e^{-j\omega\cdot 1} + e^{-j\omega\cdot 2}\right) = H(j\omega)\cdot\left(1 - 2e^{-j\omega} + e^{-j2\omega}\right). \tag{8}$$

iv) Using the convolution theorem for the DTFT evaluate $h[n]$.

$$H(j\omega) = \frac{G(j\omega)}{D_2(j\omega)} = \frac{1}{1 - 2e^{-j\omega} + e^{-j2\omega}} = \frac{1}{\left(e^{-j\omega} - 1\right)^2}. \tag{9}$$

Q4-2 A certain signal processing system operating at sampling rate T, is described by the model in Fig. 4. The system is initially relaxed, that is, $y(nT) = 0$ for $n < 0$, and p is a real constant.

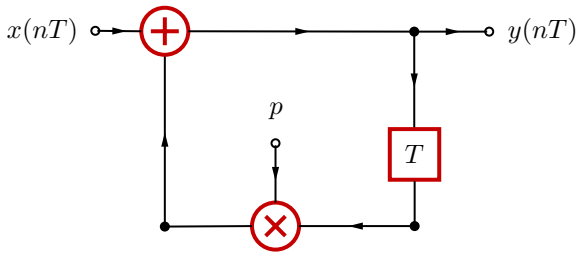


Fig. 4: System Model for Q4-3. T denotes delay by sample step T, $\otimes$ denotes *gain* or *multiplication* and $\oplus$ denotes *addition*.

(a) Find the transfer function? **[2 Marks]**

$y(nT) = x(nT) + py(nT - T)$.

(b) Find the impulse response of the system. **[2 Marks]**

With $x(nT) = \delta(nT)$, we can write $y(nT) = p^n$. Since $y(nT) = 0$ for $n \leqslant 0$, we have $y(nT) = u(nT)p^n$.

(c) Find the unit-step response ($x[n] = u[n]$) of the system. **[3 Marks]**

$y(nT) = u(nT)\frac{1-p^{n+1}}{1-p}$.

(d) Find the response of the above system to the exponential excitation, $x(nT) = u(nT)e^{\jmath\omega nT}$. **[3 Marks]**

$y(nT) = u(nT)H(e^{\jmath\omega T})(e^{\jmath\omega nT} - p^{n+1}e^{-\jmath\omega T}) = u(nT)M(\omega)(e^{\jmath(\Theta(\omega)+\omega nT)} - p^{n+1}e^{\jmath(\Theta(\omega)-\omega T)})$, where $M(\omega)$ and $\Theta(\omega)$ denotes the magnitude and phase response of the system.

Q4-3 Suppose that a certain signals is defined by ($|\mu| < 1$),

$$h[n] = \left(\sum_{|m|\leqslant n} \mu^{|m|}\right) u[n],$$

where $u[n]$ is the usual unit-step sequence. Show that its Fourier transform is given by,

$$H(e^{\jmath\omega}) = \frac{1+\mu e^{-\jmath\omega}}{(1-e^{-\jmath\omega})(1-\mu e^{-\jmath\omega})}.$$

[8 Marks]

First, notice that for $n \geqslant 0$,

$$\sum_{|m|\leqslant n} \mu^{|m|} = 2\frac{1-\mu^{n+1}}{1-\mu} - 1 = \frac{1+\mu}{1-\mu} - \frac{2\mu}{1-\mu}\mu^n.$$

Hence, h can be re-expressed as $h[n] = (\frac{1+\mu}{1-\mu} - \frac{2\mu}{1-\mu}\mu^n)u[n]$ and its z-transform is given by

$$\begin{aligned} H(z) &= \sum_{n=0}^{\infty}(\frac{1+\mu}{1-\mu} - \frac{2\mu}{1-\mu}\mu^n)z^{-n} = \frac{1+\mu}{1-\mu}\sum_{n=0}^{\infty} z^{-n} - \frac{2\mu}{1-\mu}\sum_{n=0}^{\infty}\mu^n z^{-n} \\ &= \frac{1+\mu}{1-\mu}\frac{1}{1-z^{-1}} - \frac{2\mu}{1-\mu}\frac{1}{1-\mu z^{-1}} = \frac{(1+\mu)(1-\mu z^{-1}) - 2\mu(1-z^{-1})}{(1-\mu)(1-z^{-1})(1-\mu z^{-1})} \\ &= \frac{1-\mu+\mu z^{-1}-\mu^2 z^{-1}}{(1-\mu)(1-z^{-1})(1-\mu z^{-1})} = \frac{1+\mu z^{-1}}{(1-z^{-1})(1-\mu z^{-1})}. \end{aligned}$$

Hence, its Fourier transform is given by

$$H(e^{\jmath\omega}) = \frac{1+\mu e^{-\jmath\omega}}{(1-e^{-\jmath\omega})(1-\mu e^{-\jmath\omega})}.$$